

Page 70, Complete questions #1 - #5, #11

1.

a) List four legal identifier names.

A: abc, _abc, abc1, aBc

b) List four illegal identifier names and explain why each is illegal.

A: 1abc, a b c, abc-def, abc.def

2.

a) In two statements, declare a variable named numBeads and assign it the value 5.

**A: int numBeads;
numBeads = 5;**

b) In one statement, declare a variable named numBeads and assign it the value 5.

A: int numBeads = 5;

3.

a) What is the final value of yourNumber after the last statement executes? **A: 13**

```
int myNumber = 5;  
int yourNumber = 4;  
myNumber = yourNumber * 2;  
yourNumber = myNumber + 5;
```

b) What is the final value of yourNumber after the last statement executes? **A: 16**

```
int myNumber;  
int yourNumber = 4;  
myNumber = yourNumber + 7;  
yourNumber = myNumber + 5;
```

4.

Determine the appropriate data type for each of the following values:

a) the number of basketballs in a department store. **A: int**

b) the price of a basketball. **A: double**

c) the number of players on a basketball team. **A: int**

d) the average age of the players on a basketball team. **A: double**

e) whether a basketball player has received a jersey or not. **A: boolean**

f) the first initial of a basketball player's first name. **A: String**

5.

a) What is the difference between a primitive data type and an abstract data type?

A: primitive data type: stores a single raw value (int, boolean)

abstract data type: complex data type defined by its interface (list)

b) What is the difference between a class and an object.

A: class: code template used to build objects (class Car)

object: usable copy of a class built in memory (myCar)

11.

Using the following declarations, rewrite the statements to include the appropriate type casting, rounding where necessary. If type casting is not necessary, explain why:

```
int j = 5;
```

```
double k = 1.6;
```

```
int y;
```

```
double z;
```

a) $y = j * k$;

A: $y = (int) (j * k)$;

b) $z = j * k$;

A: $z = j * k$; (type casting not necessary, the result(z) is already a double)

c) $z = k * k$;

A: $z = k * k$; (not necessary, already double)

d) $j = k$;

A: $j = (int) Math.round(k)$;

e) $k = j$;

A: $k = j$; (not necessary, int converts to double)

f) $y = j + 3$;

A: $y = j + 3$; (not necessary, both are already int)